

CASUALTY & DISASTER INSURANCE MANAGEMENT SYSTEM

1. Project Overview

A full-stack enterprise insurance platform that automates the complete lifecycle of casualty and disaster insurance — from policy creation and policy sales, through claim submission, geographic disaster verification, fraud detection, investigation, and payment. Built on .NET 8 Clean Architecture, SQL Server, and Angular with Tailwind CSS.

- ✓ Geographic disaster verification — incident location & date cross-checked against real disaster event data
- ✓ Fraud detection engine — fraud scoring engine that auto-flags suspicious claims
- ✓ 9-stage claim state tracking — not just pending/approved/rejected
- ✓ 4 specialized roles — each with distinct workflow responsibilities and data access scope
- ✓ Clean Architecture — Domain, Application, Infrastructure, API separation

Tech Stack

Layer	Technology	Purpose
Backend API	.NET 8 / ASP.NET Core Web API	RESTful endpoints, business logic, authentication
ORM	Entity Framework Core 8	DB access, Code-First migrations
Database	Microsoft SQL Server	Primary relational data store
Authentication	ASP.NET Core Identity + JWT	Secure role-based authentication
Frontend	Angular (Standalone)	SPA, reactive forms, route guards
Styling	Tailwind CSS	Utility-first component design
Architecture	Clean Architecture	Domain / Application / Infrastructure / API

2. System Roles

All five roles share a single Users table with a Role enumerator. Extended by a Customers table for customer-specific profile fields. Role-based access enforced via JWT claims on the backend and canActivate route guards on the frontend.

Role	Core Responsibilities	Claim Authority
Admin	Create policy products, manage all users, system analytics	None
Agent	Sell policies to customers, track commissions, checks whether customer is paying premium or not — linked to policy until expiry	None
Claims Officer	Review new claims, review reports + fraud detection, make final decision, trigger payment	Claim review and amount credit to customers
Customer	Buy policies, submit claims, upload documents, track claim status in portal	Submission only

Role Separation Rules

Agent is tied to the POLICY (permanent until expiry) — not to the claim. Claims officer is tied to the CLAIM (temporary, per-claim lifecycle). Admin has zero operational involvement in claims workflow.

Phase 1: Policy Management Flow

Step 1: Admin Creates Policy Product

- Admin logs in
- Creates:
 - Policy Name
 - Coverage Type (Flood, Earthquake, Fire, etc.)
 - Coverage Amount
 - Premium Structure
 - Tenure
 - Claim Limits
- Policy becomes **Available for Sale**

Output: Policy Product stored in database

Step 2: Agent Sells Policy to Customer

- customer logs in
- Selects policy
- Makes customization

- Policy gets linked:
 - To Customer
 - To Agent (permanent until expiry)

Output:

- Policy Purchase created and document will be sent to user
- Policy purchase will be notified to agent of that policy

Step 3: Customer Pays Premium

- Customer views premium schedule
- Makes payment
- System:
 - Records transaction
 - Updates payment status
 - Calculates policy active/inactive state

Policy becomes ACTIVE only if premium is paid

CLAIM LIFECYCLE WORKFLOW

This is the most critical section of your system.

You have a **9-Stage Claim State Tracking System**

Stage 1: Claim Submission

Customer:

- Selects active policy
- Enters:
 - Incident date
 - Incident location
 - Description
 - Claimed amount
- Uploads:
 - Proof documents
 - Photos
 - FIR / Government reports (if needed)

System validates:

- Policy active?
- Incident date within coverage?

- Coverage type matches?

Claim status → Submitted

Stage 2: Automatic Geographic Disaster Verification

System:

- Cross-checks:
 - Incident Location
 - Incident Date
 - Disaster Database

If:

- Disaster matches → Pass
- No disaster found → Flag

Claim status → GeoVerified / GeoFlagged

Stage 3: Fraud Detection Engine

System runs fraud scoring:

Checks:

- Multiple claims in short time?
- Same documents reused?
- Policy recently created?
- Claim amount unusually high?
- Suspicious geographic pattern?

System calculates:

FraudScore (0-100)

If score > threshold:

- Mark as High Risk

Claim status → FraudCheckCompleted

Stage 4: Assignment to Claims Officer

System:

- Auto-assigns to available Claims Officer
- Officer receives dashboard notification

Claim status → UnderReview

Stage 5: Investigation Phase

Claims Officer:

- Reviews:
 - Disaster verification result
 - Fraud score
 - Customer history
 - Documents
- Can:
 - Request additional documents
 - Approve
 - Reject
 - Escalate

If documents required:

→ Status: PendingDocuments

Stage 6: Decision Stage

Claims Officer chooses:

Approve

- Approved amount entered
- Remarks added

Status → Approved

Reject

- Rejection reason mandatory

Status → Rejected

Stage 7: Payment Processing

If approved:

System:

- Creates payment record
- Credits customer bank account
- Updates ledger

Status → PaymentProcessed

Stage 8: Claim Closure

System:

- Marks claim as Closed
- Locks further edits

COMPLETE CLAIM STATE MACHINE

Your 9-stage system:

1. Submitted
2. GeoVerified / GeoFlagged
3. FraudCheckCompleted
4. UnderReview
5. PendingDocuments
6. InvestigationCompleted
7. Approved / Rejected
8. PaymentProcessed
9. Closed

This gives your project **enterprise-level maturity**.

ROLE-BASED WORKFLOW

Admin

- Create policy products
- Manage users
- View analytics
- No claim involvement

Agent

- Sell policies
- Track commission
- View policy payment status

Agent is linked to:

Policy → Permanent

Claims Officer

- View assigned claims
- Review fraud score
- Investigate
- Approve / Reject
- Trigger payment

Claims Officer is linked to:

Claim → Temporary

Customer

- Buy policy
 - Pay premium
 - Submit claim
 - Upload documents
 - Track claim status
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Core Tables Involved

- Users
- Plan – data created by admin
- Policies (along with document) – customer will send all the details and then he will raise a buy request, application will send the final details document, now customer gets the policy and agent receives what policy is bought.
- Claims (along with documents, fraud score) – Raise a claim, claims officer will review
- Payments